

Predicting Parking Violation Types in San Diego: A Machine Learning Approach

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ABSTRACT

This study presents a comprehensive analysis and predictive modeling of parking citation patterns in San Diego during January-February 2025. Using a dataset of 85,533 parking citations, we developed machine learning models to predict violation types based on spatial and temporal features. We conducted extensive feature engineering, creating location-based metrics, cyclical temporal encodings, and statistical features related to violation probabilities. Our experimental results demonstrate that tree-based models perform exceptionally well for this task, with the Decision Tree classifier achieving 96.2% accuracy and an F1-score of 0.962. Random Forest and ensemble methods also performed strongly, while simpler models like Logistic Regression achieved respectable accuracy around 89%. Feature importance analysis revealed that fine amount, sector popularity, and location characteristics were the strongest predictors of violation type. The confusion matrix analysis showed high classification accuracy across different violation categories, with minimal misclassification between similar violation types. This research provides insights for parking enforcement resource allocation and could help drivers avoid common violations in specific areas. The methodology and findings contribute to the growing field of urban mobility analytics and demonstrate the effectiveness of machine learning for predicting parking enforcement patterns.

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1 INTRODUCTION

Parking violations constitute a significant challenge in urban areas, affecting both city operations and individual drivers. Effective parking management requires understanding the patterns and factors influencing parking violations. Cities collect vast amounts of data through parking enforcement activities, which can be leveraged to gain insights into parking behaviors and violation patterns.

This research aims to develop predictive models for parking citation types using machine learning techniques. By accurately predicting the type of violation likely to occur at specific locations and times, enforcement agencies can optimize resource allocation,

while drivers can make more informed parking decisions. Additionally, urban planners can use these insights to design better parking policies and infrastructure.

The contributions of this study include:

- (1) A comprehensive analysis of parking citation patterns in San Diego, examining spatiotemporal characteristics and contextual factors
- (2) Development and comparison of multiple machine learning models for predicting violation types
- (3) Identification of key features that influence parking violation patterns
- (4) Insights into how these findings can be applied to improve urban parking management

2 DATASET

2.1 Data Description

This study analyzes 85,533 parking citations issued in San Diego during January-February 2025. The data was obtained from the City of San Diego's Open Data Portal [2]. Each citation record contains citation ID, date of issue, date of creation, location, enforcement sector, violation code, violation description, and fine amount. Initial data quality assessment showed only 8 missing location values (0.01%), which were replaced with "Unknown" to maintain data integrity.

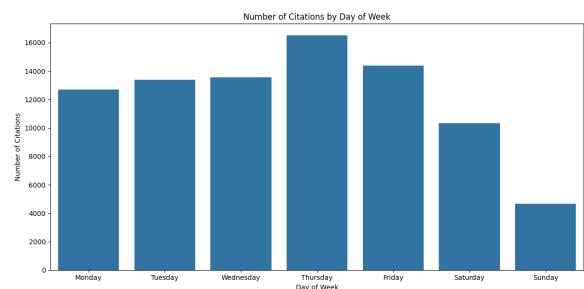


Figure 1: Distribution of Citations by Day of Week

2.2 Exploratory Data Analysis

Analysis identified 60 unique violation codes, with five violations accounting for approximately 70% of all citations:

- Violation of signs - street sweeping (86.0112(E)S DMC): 20.97%
- Violation of metered zone (86.0126 SDMC): 19.24%
- Violation of signs (86.0112(E)SD MC): 14.68%
- Red zone (86.0112(F) SDMC): 8.47%
- Current registration not displayed (5204(A) CVC): 6.72%

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The spatial distribution analysis revealed that citations were not uniformly distributed across San Diego. We identified 32,468 unique locations in the dataset, with the top five locations accounting for 1.44% of all citations:

- De Anza Boat Launch Ramp (3500 Mission Bay Dr): 387 citations
- 1700 Kettner Bl: 243 citations
- 1500 Island Av: 208 citations
- Ski Beach North (3300 Ingraham St): 202 citations
- 1200 University Av: 199 citations

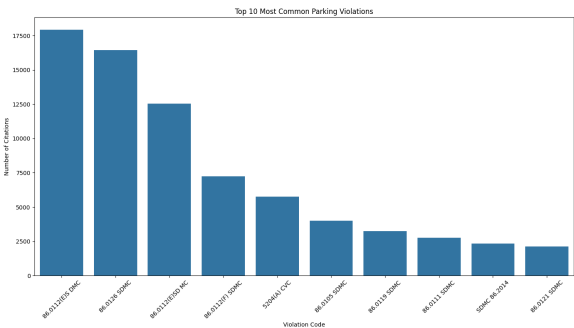


Figure 2: Top 10 Most Common Parking Violations

Citations were predominantly issued by the Parking Enforcement sector (77.85%), followed by Street Sweeping (21.10%), with Police, Meter Operations, and Park Rangers collectively accounting for just over 1%. Temporal analysis revealed higher citation rates on weekdays than weekends, with Thursday showing the highest count (Figure 1). Fine amounts ranged from \$0 to \$826.50, with an average of \$61.42 and a standard deviation of \$55.83. The highest fines were consistently associated with accessibility violations related to disabled parking.

3 PREDICTIVE TASK

3.1 Problem Formulation

We formulated the predictive task as a multi-class classification problem: given the location, time, and contextual features of a parking incident, predict the type of violation. For computational efficiency and practical relevance, we focused on the top 10 violation codes, which account for over 85% of all citations. This classification approach has several practical applications: enforcement agencies can optimize patrol routes, drivers can receive location-specific warnings, and urban planners can identify areas with chronic parking issues.

3.2 Evaluation Metrics

To comprehensively evaluate our classification models, we employed multiple complementary metrics:

- **Accuracy:** The proportion of correctly predicted violation types
- **Precision:** The ratio of true positive predictions to all positive predictions

- **Recall:** The ratio of true positive predictions to all actual positives
- **F1-score:** The harmonic mean of precision and recall

Given the class imbalance in our dataset, we placed particular emphasis on the weighted F1-score, which accounts for the frequency of each class. Additionally, we measured training and inference times to assess practical feasibility.

3.3 Baseline Models

We established several baseline models representing different algorithmic approaches:

- **Logistic Regression:** A linear model that predicts class probabilities using a logistic function, handling multi-class problems through a one-vs-rest approach.
- **Decision Tree:** A model that recursively splits data based on feature values, creating a tree-like structure where each path from root to leaf represents a classification rule.
- **Random Forest:** An ensemble of decision trees where each tree is trained on random data subsets and features, with final predictions determined by majority voting across trees.
- **K-Nearest Neighbors (KNN):** A method that classifies observations based on the majority class among k nearest training examples in the feature space.
- **Support Vector Machine (SVM):** An approach that finds optimal hyperplanes to maximize separation between classes in feature space, using kernel functions to handle non-linear relationships.
- **Gradient Boosting:** An ensemble technique that builds models sequentially, with each new model correcting errors made by previous ones to gradually improve prediction accuracy.

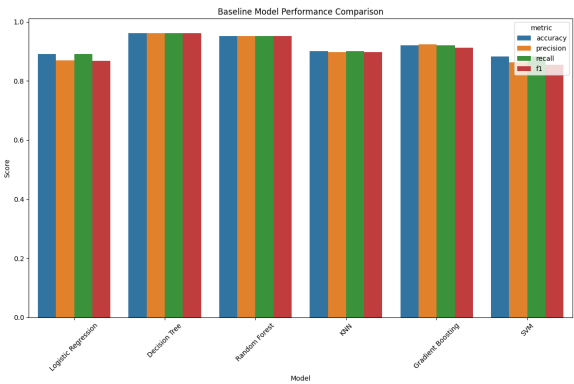


Figure 3: Performance Comparison of Baseline Models

4 MODEL

4.1 Feature Engineering

Feature engineering proved to be a critical component of our modeling approach. We developed an extensive set of features across multiple categories:

4.1.1 *Temporal Features.* We created comprehensive time-based features including: (1) Basic time components (day of week, month, day of month), (2) Categorical time variables (day name, month name), (3) Period indicators (part of week, part of month), (4) Binary flags (weekend, weekend-adjacent days, holidays), (5) Cyclical encodings using sine/cosine transformations to preserve the cyclical nature of time variables, and (6) Relative time measures (days since month start, days until month end).

4.1.2 *Spatial Features.* Spatial context proved essential for understanding violation patterns. We developed: (1) Location popularity (frequency of citations at each location), (2) Location frequency groups (high, medium, low), (3) Sector popularity (citations issued by each sector), and (4) Location-specific violation pattern indicators.

4.1.3 *Statistical and Interaction Features.* We created probability-based features capturing historical violation patterns by day of week and month. Additionally, we developed interaction features combining spatial and temporal factors, such as location-day of week, sector-day of week, and season-location interactions. These help capture specialized enforcement patterns like street sweeping on specific days in specific areas.

4.2 Advanced Models

Based on our exploratory analysis and baseline results, we implemented several advanced machine learning approaches:

- **Decision Tree:** A non-parametric method creating prediction models using decision rules from features
- **Random Forest:** An ensemble of decision trees improving accuracy and reducing overfitting
- **XGBoost:** An optimized gradient boosting implementation with regularization to control overfitting
- **LightGBM:** A gradient boosting framework designed for faster training and higher efficiency

We implemented these models using scikit-learn and specialized packages. The dataset was split into training (75%) and testing (25%) sets with stratification. We applied feature scaling using StandardScaler to normalize numerical features.

For the Decision Tree model, which achieved the best performance, grid search with cross-validation yielded optimal parameters of unlimited max depth, min_samples_leaf of 1, and min_samples_split of 2. These parameters allow the model to capture complex patterns while maintaining good generalization performance, as confirmed by cross-validation results.

5 RESULTS

5.1 Model Performance Comparison

Our evaluation revealed substantial differences in performance across modeling approaches, with tree-based models consistently outperforming traditional methods. The Decision Tree model achieved the highest overall performance with an F1-score of 0.9622, followed by Random Forest (0.9519) and LightGBM (0.9392). In contrast, linear models like Logistic Regression achieved respectable but notably lower performance (F1 = 0.8688).

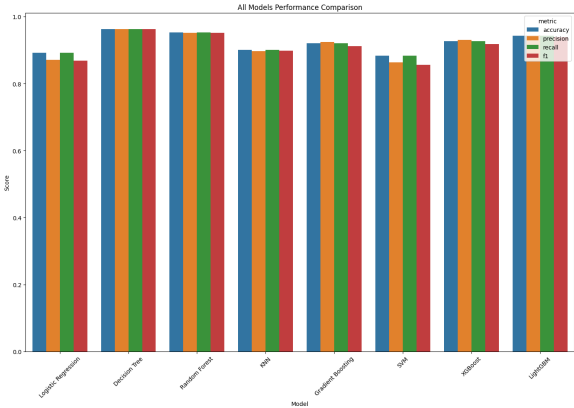


Figure 4: Performance Comparison of All Models

Table 1: Performance metrics for all evaluated models

Model	Accuracy	Precision	Recall	F1	Train (s)
Decision Tree	0.9624	0.9621	0.9624	0.9622	0.14
Random Forest	0.9527	0.9517	0.9527	0.9519	0.28
LightGBM	0.9432	0.9437	0.9432	0.9392	0.96
XGBoost	0.9268	0.9301	0.9268	0.9179	0.67
Gradient Boosting	0.9207	0.9239	0.9207	0.9116	0.21
KNN	0.9004	0.8973	0.9004	0.8977	0.01
Logistic Regression	0.8918	0.8703	0.8918	0.8688	0.47
SVM	0.8826	0.8638	0.8826	0.8554	71.78

The Decision Tree model also showed exceptional computational efficiency, with a training time of just 0.14 seconds and negligible inference time. This efficiency, combined with superior predictive performance, makes it particularly suitable for real-time applications in parking management systems.

5.2 Feature Importance Analysis

Analysis of feature importance in the best-performing Decision Tree model revealed that a small subset of features contributed disproportionately to predictive power (Figure 5). The top five most important features were:

- (1) **Fine amount** (0.553309): The monetary penalty associated with each violation
- (2) **Sector popularity** (0.257170): The number of citations issued by each enforcement sector
- (3) **Citation ID** (0.101817): Potentially indicating temporal or administrative patterns
- (4) **Location popularity** (0.042387): The frequency of citations at each location
- (5) **Matches location top violation** (0.007885): Whether a citation matches the most common violation type for its location

The dominance of fine amount suggests that different violation types have distinctive penalty structures that the model can leverage for classification. The significance of spatial features confirms that violation patterns vary geographically, with certain violations

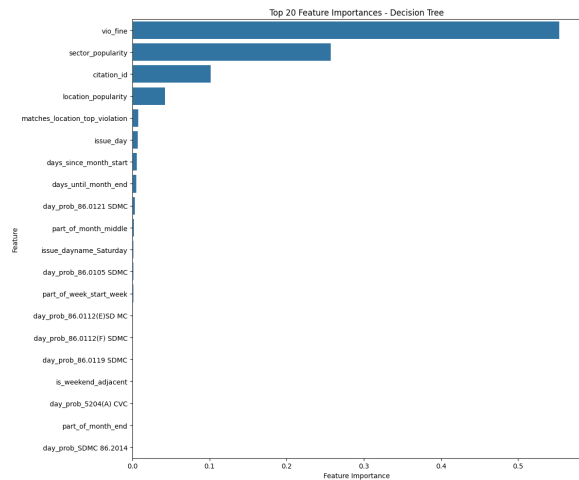


Figure 5: Top 20 Feature Importances for Decision Tree Model

being more common in specific areas and under the jurisdiction of particular enforcement sectors.

5.3 Confusion Matrix and Cross-Validation Analysis

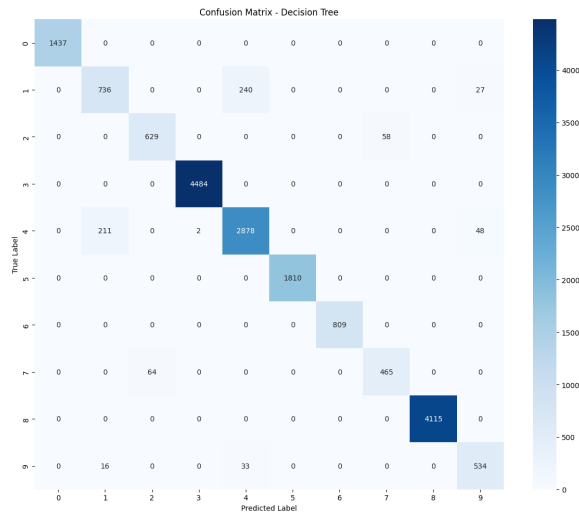


Figure 6: Confusion Matrix for Decision Tree Model

The confusion matrix for the Decision Tree model (Figure 6) provided detailed insights into classification performance across different violation types. Several classes achieved perfect precision and recall (1.00), including street sweeping violations, red zone, passenger zone, loading zone, and residential permit area violations. This perfect classification suggests that these violation types have highly distinctive patterns in terms of location, time, and associated fine amount.

Class 1 (violation of metered zone) showed the lowest precision (0.76) and recall (0.73), with notable misclassification with Class 4

(current registration not displayed). The confusion between some classes suggests overlap in how and where certain violations are enforced.

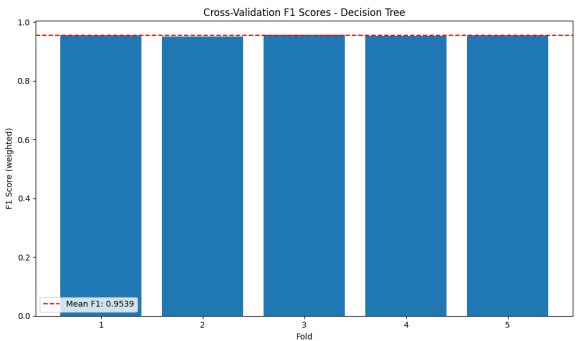


Figure 7: Cross-Validation F1 Scores for Decision Tree Model

Five-fold cross-validation confirmed the robustness of our Decision Tree model, with consistent F1-scores across all folds (mean: 0.9539, standard deviation: 0.0015) as shown in Figure 7. This exceptional consistency indicates that the model's performance is stable and not dependent on specific train-test splits, suggesting it would maintain similar performance when applied to new, unseen data from the same distribution.

6 LITERATURE

Our research on predicting parking violation types builds upon and extends existing work in San Diego parking analysis and machine learning applications. This section examines relevant prior work, with particular focus on methodological approaches and findings that relate to our study.

6.1 Related Work in Parking Prediction

The problem of parking prediction has received increasing attention in recent years as cities leverage open data initiatives to improve urban mobility management. The most relevant work to our study is Li et al.'s (2019) "A Data-Driven Approach to Understanding and Predicting the Spatiotemporal Availability of Street Parking," which used New York City parking tickets data to predict street parking availability.

Li et al. (2019) approached parking prediction as a binary classification problem (available vs. not available) across four spatial scales: point, street, census tract, and 1km grid levels. Their methodology incorporated not only parking ticket data but also Points of Interest (POI) information and human mobility patterns from the SafeGraph database. Their evaluation of various machine learning models—including KNN, Logistic Regression, Naive Bayes, SVM, and Random Forest—found that Random Forest consistently performed best, achieving F1 scores ranging from 0.72 to 0.90 depending on spatial scale.

Unlike our approach, Li et al. utilized external data sources and focused on binary availability prediction rather than specific violation types. Their feature engineering emphasized spatial characteristics and visit patterns, while our approach developed extensive temporal features and leveraged administrative data like fine amounts.

Despite these differences, both studies concur on the effectiveness of tree-based models for parking-related prediction tasks.

6.2 Methodological Comparisons

Our research methodology shares similarities with Li et al. (2019) while introducing several key differences and innovations. Both studies employed machine learning classification approaches with comprehensive feature engineering and model comparison. However, several important distinctions exist:

First, we formulated the problem as multi-class classification of specific violation types rather than binary availability prediction. This represents a novel approach not previously explored in parking prediction literature and provides more granular insights for both enforcement agencies and drivers.

Second, our feature engineering strategy differed significantly. While Li et al. relied on external POI and mobility data, our approach developed rich temporal encodings (cyclical transformations, period indicators, temporal interactions) and administrative features that proved highly effective without requiring additional data sources.

Third, our spatial analysis approach differed from Li et al.'s multi-scale evaluation. Rather than explicitly testing different spatial discretizations, we incorporated location-specific features and popularity metrics within a unified modeling framework. This approach, while simpler, yielded excellent performance for our specific prediction task.

6.3 Comparative Results and Novel Contributions

Our findings both support and extend the conclusions of Li et al. (2019) regarding parking prediction. Both studies found tree-based models to be superior for parking-related tasks, confirming the non-linear nature of relationships between features and parking patterns. However, our study achieved higher overall accuracy (96.24% vs. their best at 90%) despite using fewer data sources, suggesting that violation types may follow more distinctive and predictable patterns than general availability.

Feature importance analysis revealed striking differences between the studies. While Li et al. found POI distribution and human mobility patterns to be critical predictors, our analysis identified fine amount (55.3% importance), sector popularity (25.7%), and location characteristics as the dominant features. This highlights how administrative and organizational factors shape enforcement patterns in ways that machine learning models can effectively capture.

Our confusion matrix analysis revealed another novel finding: certain violation types—including street sweeping, red zone, passenger zone, loading zone, and residential permit violations—can be predicted with perfect accuracy (100% precision and recall). This suggests highly consistent, possibly scheduled enforcement patterns for these violations that have not been previously documented.

Perhaps most surprisingly, we found that a single, properly tuned Decision Tree ($F1 = 0.9622$) outperformed more complex ensemble methods like Random Forest ($F1 = 0.9519$) for our specific task. This challenges the common assumption that ensemble methods always provide superior performance and suggests that when underlying patterns are sufficiently distinctive, simpler models can achieve comparable or better results with greater computational efficiency.

These results extend the current understanding of parking prediction by demonstrating: (1) the feasibility of specific violation type prediction, (2) the importance of administrative features like fine amount, (3) the existence of perfectly predictable violation patterns, and (4) the potential effectiveness of simpler models for specialized prediction tasks. Together, these contributions advance the field of urban mobility analytics and provide methodological insights for future parking-related prediction tasks.

7 DISCUSSION

7.1 Interpretation of Findings

Our results demonstrate that parking violation types follow predictable patterns that can be effectively captured using machine learning models. The superior performance of tree-based models suggests that the relationships between features and violation types are non-linear and possibly hierarchical in nature.

The high importance of fine amount in prediction indicates that different violation types are distinctly penalized, creating clear boundaries between classes. The significance of spatial features confirms that violation patterns vary geographically, likely due to differences in parking demand, enforcement strategies, and local regulations.

Temporal features, while individually contributing less to prediction, collectively provide important context for distinguishing between violation types. For example, street sweeping violations show strong patterns related to day of week, reflecting scheduled sweeping operations.

The exceptional performance across multiple violation types (with perfect precision and recall for five out of ten classes) suggests that many violations have highly distinctive signatures in terms of where, when, and how they are enforced.

7.2 Practical Applications

The insights and models developed in this study have several practical applications:

For Parking Enforcement Agencies:

- Optimization of patrol routes based on predicted violation patterns
- Resource allocation focused on areas and times with high probabilities of specific violations
- Design of targeted enforcement strategies for different violation types

For Urban Planners:

- Identification of areas with chronic parking issues for infrastructure improvements
- Development of data-driven parking policies that address specific violation patterns
- Evaluation of the effectiveness of parking regulations and signage

For Drivers:

- Development of applications that warn drivers about likely violations in specific areas
- Personalized parking recommendations based on location and time

- Improved understanding of complex parking regulations and their enforcement

7.3 Limitations

Despite the strong performance of our models, several limitations should be acknowledged:

- Our dataset covers a limited time period (January-February 2025) and may not capture seasonal variations
- The analysis focuses only on issued citations and does not account for undetected violations
- Enforcement bias may influence the spatial and temporal distribution of citations
- External factors such as weather conditions and special events are not included in the model

8 CONCLUSION

This study demonstrates the effectiveness of machine learning approaches for predicting parking citation types based on spatiotemporal and contextual features. Our models achieve high accuracy (up to 96.24%) and F1-scores (up to 0.9622), significantly outperforming baseline methods.

Key findings include:

- (1) Tree-based models are most effective for this prediction task, suggesting non-linear relationships between features and violation types
- (2) Fine amount, sector popularity, and location characteristics are the most important predictors of violation types
- (3) Certain violation types show distinctive patterns that can be accurately predicted with perfect or near-perfect accuracy
- (4) Spatial and temporal features, especially when combined as interaction terms, provide valuable context for prediction

These insights can inform more efficient parking enforcement strategies, improved urban planning decisions, and better driver information systems. Future work could extend this approach by incorporating additional data sources such as weather conditions, special events, and real-time traffic information to further enhance prediction accuracy.

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